Tongyao Zhang

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Educational and Professional Experience

2026.7 – Postdoctoral Fellow, Indiana University, United States
Gates Foundation AI in Education Project, PI: Dr. Benjamin Motz

2021.1 – 2026.6 Ph.D. in Psychology, Indiana University, United States
Concentration: Developmental Psychology and Cognitive Science
Advisory committee: Dr. Emily R. Fyfe (chair), Dr. Karin H. James, Dr. Linda B. Smith, Dr. Robert L. Goldstone

2016.9 – 2020.6 B.Sc. in Psychology, Beijing Normal University, China
Concentration: Cognitive and Developmental training,
Advisors: Dr. Qingfen Hu, Dr. Roi Cohen-Kadosh, Dr. Xinlin Zhou

Research Interest

Learning and Cognitive Development, Patterning, Reasoning, AI in Education

Publications

Borriello, G. A., **Zhang**, T., Wang, W., Peng, P., & Fyfe, E. R. A meta-analysis examining relations between patterning, achievement, and cognition. (In press in *Psychological Bulletin*)

Zhang, T., Borriello, G. A., James, K. H., & Fyfe, E. R. (2025). The role of patterning skill in cognitive development and learning: A critical review. *Developmental Review, 76,* 101202. https://doi.org/10.1016/j.dr.2025.101202

Gök*, S., **Zhang***, T., Goldstone, R. L., & Fyfe, E. R. (2025). Multimedia effects in initial instruction and feedback on problem solving. *Journal of Educational Psychology, 117*(7), 1077–1094. https://doi.org/10.1037/edu0000943 (*shared first authorship)

Motz, B., Diekman, A., Goldstone, R., Green, D., Fyfe, E., Bernardini, S., …**Zhang**, T., & Murphy, M. (2025). A field-initiated vision of research infrastructure for STEM education. *International Journal of STEM Education, 12*(1), 1–13. https://doi.org/10.1186/s40594-025-00581-z

Zhang, T., & Fyfe, E. R. (2024). High variability in learning materials benefits children's pattern practice. *Journal of Experimental Child Psychology, 239*, 105829. https://doi.org/10.1016/j.jecp.2023.105829

Zhang, T., & Fyfe, E. R. The effect of combined patterning and numeracy instruction on children's mathematics learning. (Under revision in *Journal of Cognition and Development*)

Zhang, T., & Fyfe, E. R. Promoting abstract thinking with symbolic labels in pattern practice. (Under revision in *Journal of Experimental Child Psychology*)

Talks and Conference Presentations

Zhang, T., Fyfe, E. R., (expected 2026, November). Patterning Practice Tunes Learners to Focus on Structures. Poster to be presented at the Psychonomic Society 67th Annual Meeting, San Diego, CA, United States.

Zhang, T., Gök, S., Gold, G., Carvalho, P. F., Fyfe, E. R., (2026, July). Effectiveness and Efficiency of Multimedia Worked Examples vs. Practice with Feedback on Learning Problem Solving. Poster virtually presented at the Cognitive Science Society, Rio de Janeiro, Brazil.

Zhang, T., Fyfe, E. R., (2026, April). "It's always the same pattern": Promoting abstract thinking with symbolic labels in pattern practice. Poster presented at the Biennial Meeting of the Cognitive Development Society (CDS), Montréal, Canada.

Zhang, T. (2026, February). From Patterns to Numbers: The Interplay of Cognitive Skills in Early Learning. Invited talk at the Brain and Cognitive Development Lab, University of Illinois Urbana-Champaign, United States.

Zhang, T., Fyfe, E. R., (2025, June). Tuned to focus on structure: Connection between patterning and reasoning. Poster presented at the Mathematical Cognition and Learning Society (MCLS), Hong Kong S.A.R., China.

Zhang, T., Fyfe, E. R., (2025, May). Counting with patterns: The effect of including patterns in numeracy instructions. Poster presented at the Society for Research in Child Development (SRCD), Minneapolis, MN, United States.

Zhang, T., & Fyfe, E. R., (2024, June). Malleability of various patterning skills in 5- and 6-year-old children. In Nicholas A. Vest & Chen Xueliang (Chairs), Pattern learning: Empirical research about interventions, parental beliefs, and links to mathematical competence in children [Symposium]. Mathematical Cognition and Learning Society (MCLS), Washington, DC, United States.

Gök, S., **Zhang**, T., Goldstone, R., Fyfe, E. R., (2024, April). Multimedia materials are more effective in feedback than in main instruction. Poster presented at the

American Educational Research Association (AERA), Philadelphia, PA, United States.

Zhang, T., Gök, S., Fyfe, E. R., Goldstone, R., (2024, April). The impact of instructional visualizations on students' drawings during mathematical problem solving. Poster presented at the American Educational Research Association (AERA), Philadelphia, PA, United States.

Zhang, T. & Fyfe, E. R., (2024, March). Not all pattern tasks are equal: Predicting children's numeracy skills from early patterning assessments. Poster presented at the Biennial Meeting of the Cognitive Development Society (CDS), Pasadena, CA, United States.

Borriello, G. A., Peng, P., **Zhang**, T., Fyfe, E. R., (2023, March). A meta-analysis examining relations between patterning, cognition, and math and reading achievement. Poster presented at the Society for Research in Child Development (SRCD) 2023 Biennial Meeting, Salt Lake City, UT, United States.

Zhang, T. & Fyfe, E. R., (2023, June). The benefits of mixing it up: Variable materials influence children's pattern practice. Talk presented at the Midwest Cognitive Science Conference, Grand Rapids, MI, United States.

Zhang, T. & Fyfe, E. R., (2022, March). Variability in Pattern Practice. Poster presented at the Biennial Meeting of the Cognitive Development Society (CDS), Madison, WI, United States.

Peer Review

Child Development (co-review)

Teaching

2025.8 – 2025.12
Instructor of Record, Indiana University
Lab in Developmental Psychology

2024.8 – 2025.5
Associate Instructor, Indiana University
Lab in Developmental Psychology

2022.9 – 2024.5
Guest Lecturer, Indiana University
Developmental Psychology

2021.9 – present
Graduate Teaching Assistant, Indiana University
Developmental Psychology; Methods of Experimental Psychology; Lab in Developmental Psychology; Infant Development

Mentoring

2025.8 – 2026.6 Avery Refka, Ameer Abu-Salih
2025.8 – 2025.12 Willow Burton
2024.8 – 2024.10 Vigna Inturi
2024.1 – 2025.1 Nicole Namack
2023.8 – 2025.5 Ben Hanson
2023.8 – 2024.8 Allyson Utz, Jillian Pullen
2022.1 – 2023.6 Anna Settle
2022.9 – 2023.1 Rutvi Shah
2022.1 – 2022.8 Wenwen Du

Honors and Awards

2025.3 Foundry10 Travel Award for the Mathematical Cognition and Learning Society, $600

2024.6 Kempf Fellowship in Psychology, $6,000

2022.9 – 2024.8 NIH T32 Training Grant in Integrative Developmental Process, Director: Dr. Linda B. Smith, $25,000/year

2019.11 Second-Class Outstanding Undergraduate Scholarship, Beijing Normal University

2019.6 Fung Scholarship, University of Oxford (funded by Victor and William Fung Foundation), PI: Dr. Roi Cohen-Kadosh, £3,500

2018.11 Third-Class Outstanding Undergraduate Scholarship, Beijing Normal University

Memberships

American Educational Research Association (AERA)
Cognitive Development Society (CDS)
Cognitive Science Society (CogSci)
Mathematical Cognition and Learning Society (MCLS)
Society for Research in Child Development (SRCD)

Language Skills

Mandarin (Native)
English (Fluent)
Japanese (Intermediate) – JLPT N2